Fiscal Year:	FY 2014	**Task Last Updated:**	FY 04/21/2015
PI Name:	Bungo, Michael W. M.D.		
Project Title:	Cardiac Atrophy and Diastolic Dysfunction During and After Long Duration Spaceflight: Functional Consequences for Orthostatic Intolerance, Exercise Capacity, and Risk of Cardiac Arrhythmias		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77030	**Congressional District:**	18
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	99-HEDS-03
Start Date:	02/01/2004	**End Date:**	07/31/2014
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Villarreal, Jennifer	**Contact Phone:**	281-483-7306
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Flight Program:	ISS		
Flight Assignment:	ISS Increment 19 (info from PI 2/2009) NOTE: Project end date is 7/31/2014 per J. Villarreal/JSC (Ed., 10/8/14) NOTE: to be extended per D. Risin/JSC; projected new end date per PI is 9/30/2016 (Ed., 4/21/14) NOTE: End date is 3/31/2014 per JSC HRP Master Task List update 5/20/2011 (Ed., 7/21/2011) NOTE: End date corrected to 9/30/2012 per J. Dardano/JSC and PI; previous end date showed 1/10/2009 in error (1/2009 information)		
Key Personnel Changes/Previous PI:	Co-PI: Benjamin D. Levine, M.D. ; Co-PI: Michael W. Bungo, M.D. no key personnel changes occurred		
COI Name (Institution):	Levine, Benjamin (UT - Southwestern)		
Grant/Contract No.:	NNJ04HB03A		
Performance Goal No.:			
Performance Goal Text:			
	Cardiac atrophy appears to develop during spaceflight or its ground-based analogues, leading to diastolic dysfunction and orthostatic hypotension. Such atrophy also may be a potential mechanism for the cardiac arrhythmias recently identified in some crew members after long duration exposure to microgravity aboard the Mir space station. Recent work has suggested that cardiac atrophy may be progressive, without a clear plateau over at least 12 weeks of bed rest, and thus may be a significant limiting factor for extended duration space missions. The purpose of this experiment is to quantify the extent, time course, and clinical significance of cardiac atrophy and identify its mechanisms. The functional consequences of this atrophy also will be determined for cardiac filling dynamics, orthostatic tolerance at 1G and fractional G (Mars and Moon) conditions, exercise tolerance, and arrhythmia susceptibility both in space on the International Space Station (ISS), and following return to Earth. Three specific aims will be addressed:		

Task Description:	Objective 1a: To determine the magnitude of left and right ventricular atrophy associated with long duration spaceflight, and to relate this atrophy to measures of physical activity and cardiac work in-flight. Magnetic resonance imaging (MRI) will be performed pre- and post-flight as the most accurate means of measuring cardiac mass, including sufficient repetitions to document recovery. Objective 1b: To determine the time course and pattern of progression of cardiac atrophy in-flight using cardiac ultrasound. Objective 2: To determine the functional importance of cardiac atrophy for cardiac diastolic function and the regulation of stroke volume during gravitational transitions, by using innovative, non-invasive imaging techniques to measure the dynamic component of diastole including: a) Strain Rate and Tissue Doppler Imaging (TDI) to quantify ventricular wall motion with high temporal resolution; b) Color Doppler M-mode echocardiography to quantify intraventricular pressure gradients and assess the magnitude of diastolic suction; and c) MRI with myocardial tagging to quantify the rate of untwisting. Objective 3: To identify changes in ventricular conduction, depolarization, and repolarization during and after long duration spaceflight, and relate these to changes in cardiac mass and morphology (presence of fibrosis from late gadolinium enhancement and/or fat accumulation from MR spectroscopy). Although the direct assessment of arrhythmia risk in otherwise healthy subjects is imprecise, the goal of this objective is to combine the best currently available non-invasive tests of arrhythmia risk with high negative predictive value to demonstrate convincingly the absence of risk for life threaten arrhythmias. These tests will include quantifying: 1) Arrhythmia burden; 2) Delayed conduction via the signal averaged electrocardiogram (SAECG); 3) Dispersion of refractoriness from QT dispersion (QTD), QT variability and T-loop morphology; and 4) Microvolt T-wave alternans (TWA) during daily activity including routine exercise. After completion of this study, the clinical expression of cardiac atrophy during long duration spaceflight will be defined clearly, and its significance for cardiac systolic and diastolic function at rest and during gravitational transitions will be elucidated. In addition, preliminary information will be obtained regarding ventricular conduction and repolarization that will provide either strong clinical reassurance, or pathophysiologic insight into the risk for cardiac arrhythmias. Once the magnitude, time course, and inciting factors for cardiac atrophy have been determined, effective countermeasures may be applied, focused on normalizing cardiac work and volume during long-duration spaceflight. See also: http://www.nasa.gov/
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The information obtained from these spaceflight experiments also will be relevant for patients after prolonged confinement to bed rest, as well as conditions that alter cardiac stiffness such as congestive heart failure, ischemic heart disease, and normal aging.
Task Progress:	SUPPLEMENTAL REPORTING FOR FINAL REPORT (as of APRIL 2015): Data collection and analyses has been completed. Discussions of the most recent data have been presented at the NASA Investigators' Working Group held in Galveston, Texas during February 2014. A final technical report has been submitted to NASA. Further public data is not available at this time pending publication in peer review journals. Bibliography accompanying this submission describes partial results. ANNUAL PROGRESS REPORT NOVEMBER 2013: The final crewmember of our allotted cohort returned to Earth during May and completed postflight data collection by the end of June 2013. All ground-based data collections occurred nominally. Inflight data collection was hampered by minor malfunctions of the Cardiopress equipment but no significant data loss resulted. Complete data set analyses are currently underway. Abstracts will be presented at this year's Annual Scientific Session of the American Heart Association (see bibliography) and a brief report was filed with the NASA Administrator. It is planned that a major Plenary Session presentation of the data will be incorporated into the NASA Investigators' Working Group Meeting scheduled for February 2014 in Galveston, Texas.
Bibliography Type:	Description: (Last Updated: 01/12/2021)
Abstracts for Journals and Proceedings	Abdullah SM, Hastings JL, Shibata S, Hamilton DR, Thomas JD, Hughes DE, Bungo MW, Levine BD. "Effects of Prolonged Space Flight on Cardiac Structure and Function." American Heart Association Annual Scientific Session, Dallas, TX, November 16-20, 2013. Circulation 2013 Nov 26;128(22 Suppl):A18672. http://circ.ahajournals.org/cgi/content/meeting_abstract/128/22_MeetingAbstracts/A18672?sid=96b1669e-ed3a-4f2f-b3d3-71e39c54d7d2 ; accessed 4/22/15. , Nov-2013
Abstracts for Journals and Proceedings	Hastings JL, Kowal J, Daniels J, Shibata S, Platts S, Hamilton D, Page R, Ng RJ, Goldberger JJ, Bungo MW, Levine BD. "Cardiac Arrhythmias During Long Duration Spaceflight." American Heart Association Annual Scientific Session, Dallas, TX, November 16-20, 2013. Circulation 2013 Nov 26;128(22 Suppl):A18904. http://circ.ahajournals.org/cgi/content/meeting_abstract/128/22_MeetingAbstracts/A18904?sid=96b1669e-ed3a-4f2f-b3d3-71e39c54d7d2 ; accessed 4/22/15. , Nov-2013
Abstracts for Journals and Proceedings	Shibata S, Hastings JL, Abdullah SM, Bungo MW, Platts S, Hamilton DR, Levine BD. "Determinants of Cardiac Loading during Long Duration Spaceflight." American Heart Association Annual Scientific Session, Dallas, TX, November 16-20, 2013. Circulation 2013 Nov 26;128(22 Suppl):A18500. http://circ.ahajournals.org/cgi/content/meeting_abstract/128/22_MeetingAbstracts/A18500?sid=309c7fa6-5e8d-4b5b-9269-f36068e8f178 ; accessed 4/21/15. , Nov-2013
Abstracts for Journals and Proceedings	May C, Borowski A, Martin D, Popovic Z, Negishi K, Hussan, JR, Gladding P, Hunter P, Iskovitz I, Kassemi M, Bungo M, Levine B, Thomas J. "Affect of Microgravity on Cardiac Shape: Comparison of Pre- and In-Flight Data to Mathematical Modeling." ACC.14. 2014 American College of Cardiology (ACC) 2014 Scientific Sessions, Washington, DC, March 29 - 31, 2014. J Am Coll Cardiol 2014 Apr 1;63(12_S):A1096. http://dx.doi.org/10.1016/S0735-1097(14)61096-2 ; accessed 4/21/15. , Apr-2014
Abstracts for Journals and Proceedings	Negishi K, Popovic ZB, Greenberg N, Borowaski A, Martin DS, Bungo MW, Levine BD, Thomas JD. "Impact of Spaceflight on Intra-ventricular Pressure Gradient: an Interim Analysis from the Integrated Cardiovascular Study." 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. , Feb-2013
Articles in Peer-reviewed Journals	Negishi K, Borowski AG, Popovic ZB, Greenberg NL, Martin DS, Bungo MW, Levine BD, Thomas JD. "Effect of gravitational gradients on cardiac filling and performance." J Am Soc Echocardiogr. 2017 Dec;30(12):1180-8. https://doi.org/10.1016/j.echo.2017.08.005 ; PubMed PMID: 29056408 , Dec-2017
Articles in Peer-reviewed Journals	Fu Q, Shibata S, Hastings JL, Platts SH, Hamilton DM, Bungo MW, Stenger MB, Ribeiro C, Adams-Huet B, Levine BD. "Impact of prolonged spaceflight on orthostatic tolerance during ambulation, and blood pressure profiles in astronauts." Circulation. 2019 Aug 27;140(9):729-38. https://doi.org/10.1161/CIRCULATIONAHA.119.041050 ; PMID: 31319685 , Aug-2019

Articles in Peer-reviewed Journals	Khine HW, Steding-Ehrenborg K, Hastings JL, Kowal J, Daniels JD, Page RL, Goldberger JJ, Ng J, Adams-Huet B, Bungo MW, Levine BD. "Effects of prolonged spaceflight on atrial size, atrial electrophysiology, and risk of atrial fibrillation." Circ Arrhythm Electrophysiol. 2018 May;11(5):e005959. https://doi.org/10.1161/CIRCEP.117.005959 ; PMID: 29752376 , May-2018
Awards	Bungo MW, Levine BD, PIs: award accepted & presentation by Co-I Jeff Hastings at Plenary Session. "Award accepted for 'Most Compelling Results from the ISS in 2013 – Integrated Cardiovascular Results' Plenary Session. Accepted and presentation by Co-I Jeff Hastings at Plenary Session, 3rd Annual ISS Research & Development Conference of the American Astronautical Society, Chicago, June 2014." Jun-2014